Gwyneth Margaux Tangog (Gwyn)

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EDUCATION

Massachusetts Institute of Technology — **Cambridge, MA**
Masters of Engineering in Computer Science — Expected Graduation: May 2027
Bachelor of Science in Computer Science and Engineering & Mathematics — GPA: 4.8/5.0; Graduated: May 2026
Relevant Coursework: Software Construction, Design and Analysis of Algorithms, Machine Learning, Advances in Computer Vision

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INDUSTRY EXPERIENCE

Recidiviz — **June 2024 - August 2024**
Implementation Engineer Intern — Remote

- Built data ingestion pipelines and raw data views to support downstream analysis and production workflows.
- Implemented schema changes to improve data organization and accessibility across downstream systems.

The Engine Room | Github | Website — **January 2024**
Data and Tech Intern — Remote

- Performed data analysis on recent reporting data from the Light Touch Support program. Wrote automation scripts for repetitive data analysis tasks. Made and deployed an interactive data analysis dashboard.

PERSONAL PROJECTS

Pokémon TCG Pack Expected Value Calculator | Github | Website | JavaScript, HTML/CSS

- Built a web application using real Pokémon TCG card and pack data to calculate the expected value of opening booster packs.
- Implemented probability calculations to estimate the expected number of packs required to obtain a target card, allowing users to evaluate pack-opening strategies.

Personal Portfolio Website | Github | Website| Astro, HTML/CSS

- Designed and built a personal portfolio website from scratch using Astro and reusable components to showcase research, projects, and technical experience.

RESEARCH EXPERIENCE

MIT Materials Systems Laboratory — **June 2026 - current**
Graduate Research Assistant — Cambridge, MA

- Built a full-stack dashboard using Python and JavaScript to analyze material substitution across global regions using synthetic interview datasets. Github | Website

MIT Theory of Computation Group — **January 2025 - May 2026**
Undergraduate Researcher — Cambridge, MA

- Developed hardness reductions for graph threading on grid graphs and Unlimited Solo Chess.
- Designed and implemented polynomial-time algorithms to establish novel counting hardness results for Unlimited Solo Chess.
- Applied the Exponential Time Hypothesis (ETH) to establish lower bounds for polyomino tiling.

MIT Concrete Sustainability Hub — **September 2023 - June 2024**
Undergraduate Researcher — Cambridge, MA

- Developed Python data analysis and visualization pipelines for research projects on concrete carbon uptake and construction workforce retention.
- Modeled lithium price elasticity using a bottom-up demand model; contributed analysis and literature review to a peer-reviewed publication in Resources, Conservation & Recycling.

TEACHING EXPERIENCE

MIT Electrical Engineering and Computer Science Department — **September 2024 - May 2026**
Undergraduate Teaching Assistant — Cambridge, MA

- Led recitation, office hours, and class-wide review sessions for the class “Math for Computer Science” (6.1200).

Technical Skills: Python, TypeScript, JavaScript, Git, PostgreSQL, SQL, Astro, HTML/CSS
Awards: Outstanding UROP (Undergraduate Research Opportunity) Award 2024, MIT Eta Kappa Nu (HKN) Honors Society